

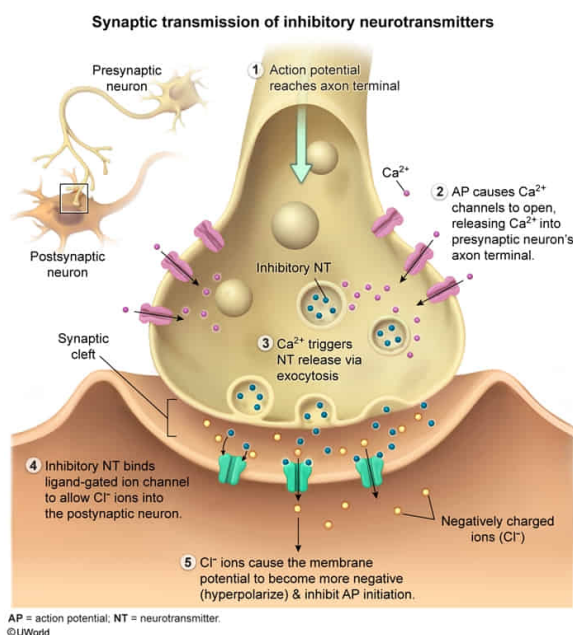
A neurotransmitter released from one neuron causes an influx of negatively charged ions into the cytosol of an adjacent neuron. In the adjacent neuron, the influx of these ions would most likely:

- ☐ A. depolarize the cell membrane and promote initiation of an action potential. [20%]
- ☐ B. hyperpolarize the cell membrane and promote initiation of an action potential. [4%]
- ☐ C. depolarize the cell membrane and inhibit initiation of an action potential. [2%]
- ✓ ☒ D. hyperpolarize the cell membrane and inhibit initiation of an action potential. [72%]

Correct

72% answered correctly

Explanation:



The vertebrate nervous system is **divided** into the central and peripheral nervous systems. Within both divisions, neurons transport electrical signals to other neurons and tissues via **action potentials** (APs).

A synapse, which can be **chemical or electrical**, is a junction formed between the axon terminal of a **pre-synaptic neuron** and the dendrites of a **post-synaptic neuron**. An AP **travels** down the axon of a pre-synaptic neuron until it reaches the axon terminals. At **chemical synapses**, the AP induces the release of **neurotransmitters** from the pre-synaptic neuron into the synaptic cleft (ie, the space between the pre-synaptic axon terminal and the post-synaptic dendrite). Neurotransmitters traverse the synaptic cleft to bind receptors on the post-synaptic membrane. This binding causes the opening of ion channels in the post-synaptic neuron, facilitating the movement of charged ions into or out of this cell. As a result, the resting **membrane potential** (-70 mV) of the post-synaptic neuron is altered. An AP is initiated in the post-synaptic neuron only if the membrane

potential becomes *more* positive and exceeds a certain threshold (-55 mV).

Depending on the neurotransmitter's effect on the post-synaptic neuron, synapses can be classified as **excitatory** or inhibitory. At an **excitatory synapse**, the pre-synaptic neuron releases neurotransmitters that cause an influx of *positively charged* ions (Na^+) into the post-synaptic neuron. These ions cause the resting membrane potential of the post-synaptic neuron to become more positive, or to **depolarize**, which *promotes* AP initiation. In contrast, **inhibitory synapses** release neurotransmitters that affect the post-synaptic neuron by causing either an influx of negative ions (Cl^-) or an **efflux of positive ions** (K^+). The *exit* of positive ions or *entry* of negative ions into the post-synaptic neuron causes the cell's membrane potential to become *more* negative, or to **hyperpolarize**, which *inhibits* AP initiation.

In this scenario, a neurotransmitter released from one neuron causes an influx of negatively charged ions into an adjacent (post-synaptic) neuron. This influx would cause the adjacent neuron's membrane potential to become *more* negative, **hyperpolarizing the post-synaptic membrane potential** and **inhibiting AP initiation (Choices A, B, and C)**.

Educational objective:

At inhibitory synapses, pre-synaptic neurons release neurotransmitters that cause either an influx of negative ions into the post-synaptic neuron or an efflux of positive ions out of the post-synaptic neuron. This causes hyperpolarization of the membrane potential and inhibits action potential initiation in the post-synaptic neuron.

Time spent:0 sec

Subject:
Biology

QID:402493

System:
Endocrine and Nervous Systems